Kalman ~~filter~~ Filter:

1. Linear stochastic system:

$$\left.\begin{aligned} x_k &= F_k x_{k-1} + B_k u_k + w_k \\ z_k &= H_k x_k + n_k \end{aligned}\right\}$$

z_k ↓ sensor outputs a value that is in Linear with true value + noise.

x_k : unknown true state.

- u_k : known input.
- F_k : state-transition matrix. (known)
- B_k : input matrix. (known)

z_k : sensor measurement.

H_k : measurement matrix

- w_k : process noise
- n_k : measurement noise.

$\}\rightarrow$ $w_k \sim N(0, Q_k)$ (independent).
$n_k \sim N(0, R_k)$.

(mean ↓, standard dev. ↓)

the filter seores $\hat{x}_k$ (estimated state) & P_k.

P_k = estimation-error covariance.

$$P_k = E\left[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T\right].$$

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2. Covariance.

$E(x) \rightarrow$ mean of $x = \frac{1}{N}\sum_{i=1}^{N} x_i$

Variance: $Var(X) = E[(x_i - \bar{x})^2] \approx \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$

(square of stdev)

covariance: $cov(X, Y) = E[(X - \mu_x)\cdot(Y - \mu_y)]$

for sampled data:

$$cov(X, Y) = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y}).$$

covariance matrix: $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow P = \begin{bmatrix} Var(x_1) & Cov(x_1, x_2) \\ Cov(x_2, x_1) & Var(x_2) \end{bmatrix}$ Symmetric.

$P = P^T$

3. prediction of state mean.

suppose after previous measurement correction:

$$\hat{X}_{k-1}^{+} = E[X_{k-1}].$$

$\because$ $X_k = F_k X_{k-1} + B_k u_k + w_k$.

take E[] on both:

$$E[X_k] = E[F_k X_{k-1} + B_k u_k + w_k].$$

$\because$ F_k, B_k, u_k are known. (fixed at each k).

$\therefore$ $E[X_k] = F_k \cdot E[X_{k-1}] + B_k E[u_k] + E[w_k]$.

since $w_k \sim N(0, Q_k)$, so $E[w_k] = 0$.

$\therefore$ $E[X_k] = F_k E[X_{k-1}] + B_k E[u_k]$.

$\because$ ~~$X_k = \hat{X}_k$~~ also $E[X_{k-1}] = \hat{X}_{k-1}^{+}$

So $\boxed{\hat{X}_k^{-} = F_k \cdot \hat{X}_{k-1}^{+} + B_k u_k}$. estimation before next measurement correction.

next state prediction before measurement correction = $F_k \cdot$ (previous state prediction after measurement correction) $+ B_k u_k$.

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predicted estimation error:

$$\tilde{X}_k^{-} = X_k - \hat{X}_k^{-}$$

true state: $X_k = F_k X_{k-1} + B_k u_k + w_k$.

$$\tilde{X}_k^{-} = F_k X_{k-1} + B_k u_k + w_k - \hat{X}_k^{-}$$

$$= F_k X_{k-1} + B_k u_k + w_k - F_k \hat{X}_{k-1}^{+} - B_k u_k.$$

$\boxed{\tilde{X}_k^{-} = F_k (X_{k-1} - \hat{X}_{k-1}^{+}) + w_k.}$ prediceed error.

dicted covariance :

$$P_k^- = E[\tilde{x}_k^-(\tilde{x}_k^-)^T].$$

substitute error : $\tilde{x}_k^- = F_k(x_{k-1} - x_{k-1}^+) + w_k.$

$$P_k = E[(F_k x_{k-1} - x_{k-1}^+ + w_k)(F_k x_{k-1} - x_{k-1}^+ + w_k)^T]$$

since $\tilde{x}_k^- = F_k(x_{k-1} - x_{k-1}^+) + w_k.$

$$\boxed{\tilde{x}_k^- = F_k \cdot (\tilde{x}_{k-1}^+) + w_k.}$$ prediction error before measurement correction.

so $P_k = E[(F_k \cdot (\tilde{x}_{k-1}^+) + w_k) \cdot (F_k(\tilde{x}_{k-1}^+) + w_k)^T].$

inside E •, expanding :

$$F_k \tilde{x}_{k-1}^+ (F_k(\tilde{x}_{k-1}^+) + w_k)^T + w_k^T(F_k(\tilde{x}_{k-1}^+) + w_k)$$

$$= F_k \cdot \tilde{x}_{k-1}^+ \cdot F_k^T \cdot \tilde{x}_{k-1}^{+T} + F_k \tilde{x}_{k-1}^+ w_k^T + w_k^T \cdot F_k \cdot \tilde{x}_{k-1}^+ + w_k^T \cdot w_k.$$

so $P_k = E(F_k \cdot \tilde{x}_{k-1}^+ \cdot F_k^T \cdot \tilde{x}_{k-1}^{+T}) \rightarrow F_k E[\tilde{x}_{k-1}^+ \cdot \tilde{x}_{k-1}^{+T}] \cdot F_k^T = F_k P_{k-1}^+ F_k^T$

$+ E(F_k \tilde{x}_{k-1}^+ w_k^T) \rightarrow F_k E[\tilde{x}_{k-1}^+ w_k^T] \rightarrow 0$

$+ E(w_k^T \cdot F_k \tilde{x}_{k-1}^+) \rightarrow 0$ (previous state error and current state process noise are independent)

$+ E(w_k^T \cdot w_k) \rightarrow Q_k.$

so $\boxed{P_k = F_k P_{k-1}^+ F_k^T + Q_k.}$ → process noise covariance.

- P_k → next predicted covariance
- F_k → physical model
- P_{k-1}^+ → prior predicted covariance

5. Derivation of the innovation (measurement error).

$$z_k = H_k x_k + n_k.$$

filter predicts that: $\hat{z}_k = H_k \cdot \hat{x}_k^-$

$$y_k = z_k - H_k \hat{x}_k^-$$

(difference between measurement and prediction).

$$y_k = H_k x_k + n_k - H_k \hat{x}_k^- = H_k (x_k - x_k^-) + n_k = H_k \tilde{x}_k^- + n_k.$$

covariance: $S_k = E[y_k y_k^T]$

$$= E[(H_k \tilde{x}_k^- + n_k)(H_k \tilde{x}_k^- + n_k)^T].$$

Similarly to P_k; since predicted state error and measurement noise are independent,

also $E[n_k^T n_k^T] = R_k$.

So $\boxed{S_k = H_k P_k^- H_k^T + R_k.}$

6. Linear correction.

$$\hat{x}_k^+ = \hat{x}_k^- + k_k y_k = \hat{x}_k^- + k_k (z_k - H_k \hat{x}_k^-)$$

(kalman gain)

since $\tilde{x}_k^+ = x_k - \hat{x}_k^+ \rightarrow \tilde{x}_k^+ = x_k - \hat{x}_k^- - k_k y_k.$

since $\begin{cases} \tilde{x}_k^- = x_k - \hat{x}_k^- \\ y_k' = H_k \tilde{x}_k^- + n_k \end{cases}$

So $x_k^+ = x_k - x_k + \hat{x}_k^- - k_k (H_k \tilde{x}_k^- + n_k) = (I - k_k H_k) \tilde{x}_k^- - k_k n_k.$

substitute into P_k with x_k^+:

$$\boxed{P_k^+ = E[\tilde{x}_k^+ \tilde{x}_k^{+T}] = (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T}$$

7. find optimal ~~covariance~~ kalman gain.

trace • : $tr(A) = \sum_i a_{ii}$

for covariance matrix : $tr(P) = E[\tilde{x}_1^2] + E[\tilde{x}_2^2] + \cdots$

total square error : $\|\tilde{x}\|^2 = \tilde{x}_1^2 + \tilde{x}_2^2 + \cdots$

So $tr(P) = E[\|\tilde{x}\|^2]$ → optimizing $tr(P)$ is • minimizing total square error.

$P^+ = (I - KH) P^- (I - KH)^T + KRK^T$ (assume at timestamp k).

$= (P^- - KHP^-)(I - H^TK^T) + KRK^T$

$P^+ = P^- - KHP^- - P^-H^TK^T + KHP^-H^TK^T + KRK^T$.

$P^+ = P^- - KHP^- - P^-H^TK^T + K(HP^-H^T + R)K^T$

let $S = HP^-H^T + R$.

then $P^+ = P^- - KHP^- - P^-H^TK^T + KSK^T$

$tr(P^+) = tr(P^-) - tr(KHP^-)$ since $tr(A) = tr(A^T)$

So $tr(P^+) = tr(P^-) - 2\,tr(KHP^-) + tr(KSK^T)$

let $J(k) = tr(P^+)$

$\frac{\partial J}{\partial K} = -2HP^- + 2KS. \Rightarrow 0.$

$2KS = 2HP^- \rightarrow$ $\boxed{K = P^-H^TS^{-1}}$

~~$K = SHR$~~ where $S = HP^-H^T + R$

So $\boxed{K = HP^-(HP^-H^T + R)^{-1}}$

Now with optimal Kalman Gain.

$$\begin{cases} P^+ = (I-KH)P^-(I-KH)^T + KRK. \\ K = P^-H^T(HP^-H^T+R)^{-1} \end{cases}$$

$$P^+ = P^- - KHP^- - P^-H^TK^T + K(HP^-H^T+R)K^T$$

$$P^-H^T(S^{-1}) \cdot S \cdot K^T = P^-H^TK^T$$

$$P^+ = P^- - KHP^- - P^-H^TK^T + P^-H^TK^T = P^- - KHP^- = (I-KH)P^-$$

so $\boxed{P^+ = P^- - P^-H^T(HP^-H^T+R)^{-1}P^-}$ simplified covariance.

Summary of Kalman Filter Algorithm:

$$\hat{X}_k^- = F_k\hat{X}_{k-1}^+ + B_ku_k. \quad \rightarrow \text{estimation \& prediction of state.}$$
$$P_k^- = F_kP_{k-1}^+F_k^T + Q_k. \quad \rightarrow \text{combine with last prediction's covariance. (confidence).}$$

measurement error

$$y_k = z_k - H_k\hat{X}_k^- \quad \leftarrow \text{measurement}$$
$$S_k = H_kP_k^-H_k^T + R_k$$

, $z_k = H_kx_k + n_k$.

$$K_k = P_k^-H_k^TS_k^{-1}$$ find optimal kalman gain

$$\hat{X}_k^+ = \hat{X}_k^- + K_ky_k$$ correct prediction

$$P_k^+ = (I-K_kH_k)P_k^-(I-K_kH_k)^T + K_kR_kK_k^T.$$ Calculate covariance (confidence).

Q_k: process noise stdev

R_k: measurement noise stdev.